Renormalization and Imprint Filtering: Source Identification, Memory, and Calibration

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Abstract

A physical trace is readable relative to a source question and a permitted measurement. We make that relation operational on a continuing lattice field: bounded randomized forces at two durations separate an unknown source relaxation rate from an unknown positive reader gain, and the acquired response history selects the next read duration and physical force. The changed carrier continues through calibration and action. A finite martingale certificate retains its nonstationary contribution and tolerates unknown centered independent read noise with a known variance cap. Exact Gaussian block spectra establish the scale-dependent distinctions on which this inquiry acts. Coarse readings can retain mass information while suppressing higher-gradient contrasts, and mixing relaxation rates makes retained history useful for prediction. Independent preparations give separate gain-free decision laws and explicit low-gain budget obstructions. History also changes which reading is acquired: five retained bits beat every fixed allocation in a slowly changing source family. On one field trajectory, a paid block reading or no new reading is followed by a bounded force. Gaussian conditioning gives the optimal one-round rule, and sixteen-bit records preserve its strict advantage over every history-independent reading choice in the stated case. The noise extension concerns response calibration; the Gaussian likelihood and conditional-state results keep their original noise assumptions. Reading prices, mechanical switching work and the supplied apparatus remain separate parts of the resource account.

1 Which traces survive a change of scale?

The world can carry a history without presenting that history in a form an observer can use. Large-scale fluctuations may identify a material's phase while revealing little about its microscopic couplings. A present measurement may be insufficient to predict the next one even when the physical dynamics is Markovian. A changing signal may reflect either the source or the instrument. These are different failures of readability.

The imprint proposal asks how generative conditions remain legible in the structures they produce [6]. Renormalization makes scale central to this question: an effective description preserves selected distinctions and suppresses others. Information geometry already relates coarse-graining to parameter distinguishability [7, 8]. Projection methods

likewise establish that eliminating degrees of freedom can introduce memory [9]. We use these ideas together, with the reading experiment and its resources specified.

Our central claim is that an imprint is a relation among a source contrast, its surviving carrier, and a reading interface. It is not an additional substance carried by every organized object. A single lattice model makes the relation calculable. Its static distribution gives source-identification tests; its dynamics gives a value for retained history; an uncertain gain gives a calibration problem. Information can survive physically without being readable through a given interface, and successful world-reading can require a model of the reader's own coupling.

Block averaging is an exact transformation of the Gaussian probability law, not a claim that the effective action stays within a short list of local couplings. The mathematical tools are standard. Their common implementation establishes what scale filtering, memory, and calibration jointly imply for the imprint account. In particular, records can serve as references for one another through either spatial scale or controlled readout settings. Whether this removes an unknown gain depends on the actual record distribution, not just on a cost assigned to each reading.

The initial calculations use three experiments in the same model family: independent static preparations, the subsequent relaxation of a preparation in a common bath, and prediction along a stationary trajectory. Their inputs are stated separately. They are not claimed to be simultaneous observations of one run. Section 9 adds a fourth experiment: a persistent regime changes which of two scales carries the current source contrast. Its retained history changes the next allocation of real readings. This is not information recovered by postprocessing a discarded field. Section 10 instead follows one field through past observation, a selected current reading and a subsequent force. Its control changes the carrier that will be read again. Section 11 uses that actual change as a calibration reference. It separates source dynamics from reader gain on one spectral mode, then uses the response history to choose a future temporal scale and control. These are declared protocols on one lattice family, not simultaneous free access to every observable.

2 A source family and its block observations

On $\Lambda = (\mathbb{Z}/L\mathbb{Z})^2$, take a real centered Gaussian field with unitary Fourier convention $\widehat{X}(k) = L^{-1}\sum_x e^{-ik\cdot x}X_x$ and precision

$$\begin{aligned} K_\theta(k) &= \mu + s(k) + g\,s(k)^2 + \gamma\,a(k), \\ s(k) &= 4\sin^2(k_1/2) + 4\sin^2(k_2/2), \\ a(k) &= 16\sin^4(k_1/2) + 16\sin^4(k_2/2). \end{aligned} \tag{1}$$

Here $\mu > 0$, $g, \gamma \geq 0$ suffice for positivity. Derivatives at $g = \gamma = 0$ use the open finite-lattice extension where $K_\theta > 0$. No claim about efficient estimation at a parameter-space boundary is needed. The parameters describe preparation conditions, not every detail of a sample's formation history.

The coefficients g and γ change higher-gradient terms. For example, $\gamma = 1/12$ cancels the fourth-order anisotropic small-momentum error of the nearest-neighbor Laplacian.

For $\ell \mid L$, observe nonoverlapping means $Y_j = \ell^{-2}\sum_{a_1,a_2=0}^{\ell-1} X_{\ell j+a}$. There are $N_\ell = (L/\ell)^2$ coarse coordinates. For coarse momentum Q, its aliases are $k_n = (Q + 2\pi n)/\ell$, with $n \in \{0,\dots,\ell-1\}^2$. Set $W_\ell(k) = \sum_{a=0}^{\ell-1} e^{ika}$.

Proposition 2.1 (Exact block spectrum)**.** *The unitary Fourier covariance of the coarse field has eigenvalues*

$$v_{\theta,\ell}(Q) = \ell^{-6} \sum_{n} \frac{|W_\ell(k_{n,1}) W_\ell(k_{n,2})|^2}{K_\theta(k_n)}. \tag{2}$$

Proof. Insert the inverse fine Fourier transform in the block mean and take the coarse unitary transform. The coarse-site sum selects the alias set. Each fine coefficient is multiplied by $\ell^{-3} W_\ell(k_{n,1}) W_\ell(k_{n,2})$. Squaring and summing its variance gives (2). □

For white noise $K = 1$, this yields $v_{\theta,\ell} = \ell^{-2}$. The zero coarse Fourier mode has variance $1/(\ell^2\mu)$; the literal global spatial mean has variance $1/(L^2\mu)$. Both are blind to g and γ. Nontrivial fluctuations can therefore be completely uninformative about a specified source contrast.

Fourier coefficients of a real field occur in conjugate pairs. Information sums below count the corresponding independent real sine and cosine coordinates, not independent complex observations twice. The effective precision is the inverse coarse covariance. It generally contains terms beyond (1); we do not assume a closed three-parameter RG flow.

3 Information retained by a coarse description

Definition 3.1 (Readable source contrast)**.** *A source contrast is readable through a fixed protocol if its two preparations induce different record distributions. Its strength can be expressed by decision risk or divergence. A resource claim also states how the records are acquired, retained, and processed.*

Proposition 3.2 (Gaussian information)**.** *One complete coarse-field snapshot has Fisher information*

$$F_{ij}^{(\ell)} = \frac{1}{2} \sum_{Q} \partial_i \log v_{\theta,\ell}(Q)\, \partial_j \log v_{\theta,\ell}(Q). \tag{3}$$

For two preparations with spectra v_0, v_1, its Chernoff information is

$$C_\ell = \max_{0 \le u \le 1} \frac{1}{2} \sum_{Q} \left[u \log v_0 + (1-u) \log v_1 + \log\left(\frac{u}{v_0} + \frac{1-u}{v_1} \right) \right]. \tag{4}$$

It is zero exactly when the two coarse covariances coincide.

Proof. In real Gaussian coordinates the variance score is $\frac{1}{2}\partial_i \log v\,(Z^2/v - 1)$; its second moment gives (3). Integrating $p_0^u p_1^{1-u}$ gives (4). Equality at zero requires equal Gaussian laws. □

A zero Fisher direction at one parameter value means only first-order local indistinguishability. It need not describe a globally unobservable source family. The zero mode above, by contrast, is exactly blind to all g, γ contrasts at fixed μ.

Nested block averages of the same sample satisfy

$$F^{(\ell_2)} \preceq F^{(\ell_1)}, \qquad C_{\ell_2} \le C_{\ell_1} \quad (\ell_1 \mid \ell_2).$$

The coarse score is the conditional expectation of the finer score; conditional variance proves the first inequality. Conditional Hölder inequality proves the second. Known parameter-independent coordinate or amplitude rescaling changes neither information. Acquiring a different observable or preparing more fields changes the experiment and is not postprocessing of the old records.

4 Reading cost and spectral access

Spatial block values are correlated. Thus $F^{(\ell)}/N_\ell$ and C_ℓ/N_ℓ are densities for a complete snapshot, not information in arbitrary independent block readings. For n independently prepared snapshots, the equal-prior optimal error obeys

$$P_e \le \tfrac{1}{2} e^{-nC_\ell}. \tag{5}$$

A 5% sufficient certificate needs $\lceil \log(10)/C_\ell \rceil$ snapshots and N_ℓ times as many scalar records. Correlated snapshots from one trajectory cannot replace independent preparations.

Table 1 uses $L = 256$, $\theta_0 = (.01, 0, 0)$. The mass contrast changes μ to .0125; the microscopic contrast changes γ to 1/12. This table idealizes continuous precision.

ℓ	N_ℓ	$F_{\mu\mu}$	$F_{\gamma\gamma}$	C_{mass}	C_{micro}
1	65536	262732.0	248831.1	.183030	170.010
4	4096	256284.8	433.963	.178014	.333084
8	1024	242440.5	8.50597	.167361	.00671261
16	256	202231.1	.133228	.137232	.000107754
32	64	126311.3	.00206272	.0831354	.00000168534

Table 1: Information in one complete block snapshot.

For the mass contrast, changing $\ell = 1$ to 32 reduces the sufficient record count from 851968 to 1792, while increasing preparations from 13 to 28. For the microscopic contrast, preparations grow from 1 to 1366242. The former is a trade between preparation and readout, not information gained by discarding data. Preparation count, physical support, reads, precision, retention, and computation belong in separate resource entries.

Changing spectral access gives another comparison. For the microscopic contrast, a real coordinate at $k = (\pi, 0)$ has variances .24937656 and .18714910. Use 255 fixed thresholds equally spaced in $[-8\sqrt{v_{\max}}, 8\sqrt{v_{\max}}]$, giving 256 outcomes including the two unbounded tails. Integrating the Gaussian laws over these bins gives Chernoff information .00514034 per independent reading. Then 448 independent preparations suffice for the 5% certificate, with 3584 bits of recorded outcomes. Counters, decoding, preparation, and the full spatial support of the spectral observable are additional resources. One recorded number is not a single-site interaction.

The low coordinate $k = (2\pi/256, 0)$ instead has continuous Chernoff information about 5.08×10^{-13}; the zero mode has none. A new interface reveals a distinction by accessing the carrier again, not by reconstructing discarded data from the old coarse records.

5 Reading a past preparation after relaxation

Prepare $X_0 \sim N(0, K_\theta^{-1})$ and subsequently use a common bath with fixed $K_0 = K_{\theta_0}$, independent of the preparation parameter θ:

$$dX_t = -K_0 X_t \, dt + \sqrt{2} \, dW_t. \tag{6}$$

Physical loss of preparation history and spatial coarse-graining are now distinct operations in the same process.

Proposition 5.1 (Remaining preparation information)**.** *The fine variance at time t is*

$$V_\theta(k,t) = K_0(k)^{-1} + e^{-2K_0(k)t}[K_\theta(k)^{-1} - K_0(k)^{-1}]. \tag{7}$$

Replace K_θ^{-1} by V_θ in (2) for its block spectrum. At $\theta = \theta_0$, its log-variance derivative is

$$r_i(Q,t) = -\frac{\sum_n |W_1 W_2|^2 e^{-2K_0 t}(\partial_i K_\theta)/K_0^2}{\sum_n |W_1 W_2|^2/K_0}. \tag{8}$$

Proof. The solution is $e^{-K_0 t}X_0 + \sqrt{2}\int_0^t e^{-K_0(t-s)}dW_s$. Its independent covariance contributions give (7). Differentiate and project to obtain (8). □

The source parameter is no longer being applied after preparation. For $K_0(k) \geq \mu > 0$, its full-field Fisher matrix at the reference satisfies

$$F_{\text{full}}(t) \preceq e^{-4\mu t} F_{\text{full}}(0).$$

Each mode contributes a positive rank-one matrix multiplied by $e^{-4K_0(k)t}$. A fixed reader adds further loss; this formula does not compare every coarse reader against its own initial information.

In the reference field of Table 1, full-field γ information falls from 248831.1 at $t = 0$ to 49.9307 at $t = 1$ and .0237407 at $t = 10$. Mass information at $t = 10$ remains 101633.8. A readable history has a depth relative to its source question, not a single universal lifetime. The bath and elapsed time are known inputs.

6 The prediction value of a retained record

Write $K = K_0$ and start (6) in its stationary law $N(0, K^{-1})$. For a real coarse Fourier coordinate $Y(t)$ of variance $v > 0$, the normalized correlation is

$$r(t) = \sum_n p_n e^{-K(k_n)|t|}, \qquad p_n = \frac{|W_1 W_2|^2/K(k_n)}{\sum_m |W_1 W_2|^2/K(k_m)}. \tag{9}$$

Theorem 6.1 (Memory relative to the coarse interface)**.** *Fix $\Delta > 0$, write $Y_j = Y(j\Delta)$, $r_1 = r(\Delta)$ and $r_2 = r(2\Delta)$. Then*

$$d = r_2 - r_1^2 = \mathrm{Var}_p(e^{-K\Delta}) \geq 0. \tag{10}$$

The minimum square risks for predicting Y_1 from Y_0 and from (Y_0, Y_{-1}), over all measurable predictors, are

$$R_1 = v(1 - r_1^2), \qquad R_2 = R_1 - \frac{vd^2}{1 - r_1^2}. \tag{11}$$

The improvement is strict exactly when the positive-weight alias set has more than one relaxation rate.

Proof. Equation (10) follows from the positive mixture (9). Positivity of K and Δ gives $0 < r_1 < 1$. The three records are jointly Gaussian with correlations r_1, r_1, r_2. Their conditional expectation minimizes square loss among all measurable functions. Its Schur-complement variance gives (11); the minimizing two-record predictor is

$$\widehat{Y}_1 = a_1 Y_0 + a_2 Y_{-1}, \qquad a_1 = \frac{r_1(1 - r_2)}{1 - r_1^2}, \quad a_2 = \frac{d}{1 - r_1^2}. \tag{12}$$

A mixture variance vanishes exactly when the rates on its support agree. □

This also characterizes Markov closure. Distinct coarse real Fourier coordinates are independent processes. If each alias group has one rate, every coordinate is an Ornstein–Uhlenbeck process. Otherwise (10) gives conditional dependence of past and future after conditioning on the entire current coarse field, so those variables alone are not Markov. The complete microscopic state remains Markov.

For $L = 64$, $\ell = 4$, $\mu = .1$, $g = \gamma = 0$, coarse Fourier index $(3, 5)$ and $\Delta = .2$, the results are

$$v = .10868306, \quad d = .005993705, \quad R_1 = .0214457374, \quad R_2 = .0214259507.$$

The improvement is .0923%.

6.1 Finite storage and arithmetic

Corollary 6.2 (Finite record precision)**.** *Quantize each record to 2^B uniform levels in $[-A, A]$, clipping outside. Put*

$$\delta = \frac{2A}{2^B - 1}, \quad z = A/\sqrt{v}, \quad \eta = \delta/2 + \sqrt{2v[z\phi(z) + \overline{\Phi}(z)]},$$

where $\phi, \overline{\Phi}$ are the standard normal density and upper tail. Applying (12) *to these records has risk*

$$\widetilde{R}_2 \le R_2 + (|a_1| + |a_2|)^2\eta^2. \tag{13}$$

If the right side is below R_1, this finite-record predictor beats every predictor using the exact current record alone.

Proof. The quantization error has L^2 norm at most η, by the Gaussian tail identity $\mathbb{E}[Y^2\mathbf{1}_{\{|Y|>A\}}] = 2v[z\phi(z) + \overline{\Phi}(z)]$. The conditional-prediction residual is independent of both records and hence orthogonal to their quantization errors. Its risk adds to the predictor perturbation's second moment. The L^2 triangle inequality then gives (13). □

In this example, $B = 10$ and $A = 8\sqrt{v}$ give $\eta \le .00257817$ and $\widetilde{R}_2 \le .021431324 < R_1$. Only the retained data occupy 20 bits. The full-support reading operation, model coefficients, and computation still have costs.

The rational-arithmetic construction in Appendix B retains this strict prediction advantage with finite coefficient and output registers.

The mechanism is different from asserting that every viable system needs memory. Coarse observation removes the sufficiency of the present. A past record then supplies information missing from that present, and its predictive value can be computed.

7 Source identification and instrument calibration

The previous calculations assumed a known reading interface. Now take $K(k) = \mu + s(k)$ and an unknown fixed positive instrument gain b. A real spectral reading has law

$$Z_i \sim N\left(0, \frac{b^2}{\mu + \lambda_i}\right), \qquad \lambda_i = s(k_i).$$

The same gain acts at both known frequencies. A positive sign is stipulated because centered variance data cannot determine it.

Proposition 7.1 (Source–gain separation)**.** *One spectral band cannot identify μ, b separately. Two bands with $\lambda_1 \neq \lambda_2$ identify both globally. For $n_1, n_2 > 0$ independent preparations, let $\eta_b = \log b$ and $q_i = -1/(\mu + \lambda_i)$. The Fisher matrix and its Schur complement are*

$$F = \frac{1}{2} \sum_i n_i \begin{pmatrix} q_i \\ 2 \end{pmatrix} \begin{pmatrix} q_i & 2 \end{pmatrix}, \tag{14}$$

$$F_{\mu|\eta_b} = \frac{n_1 n_2}{2(n_1 + n_2)} (q_1 - q_2)^2 > 0. \tag{15}$$

At a fixed total reading count the latter is maximized by equal allocation.

Proof. One variance determines only $b^2/(\mu + \lambda_i)$. For two population variances set $\rho = v_1/v_2$. Then

$$\rho = \frac{\mu + \lambda_2}{\mu + \lambda_1}, \qquad \mu = \frac{\lambda_2 - \rho\lambda_1}{\rho - 1}, \qquad b^2 = v_1(\mu + \lambda_1).$$

This determines the positive gain uniquely. Apply (3) to the log variances and take a Schur complement to obtain (15). For fixed $n_1 + n_2$, their product is maximal at equal allocation. □

For $L = 64$, $\mu = .1$, modes $(10\pi/64, 0)$ and $(\pi/2, 0)$, and 64 preparations per band, $F_{\mu|\eta_b} = 99.88849656$. Population identifiability and finite-sample information do not assert that a sample-variance ratio is unbiased or attains an information bound.

The spectral family, band locations, gain stability, and common frequency response are inputs. Their importance is physical: a system can distinguish source change from gain change because these alternatives leave different joint records. To exercise that capacity, it must estimate, retain the result, and revise subsequent reading or action. T-DOME III and Foundation III provide separate drift and feedback constructions for those operations [12, 13]. The existence of an identifiable model is not itself an agent that uses it.

The continuing experiment in Section 11 turns that distinction into the next operation. A fresh force coin supplies an intervention reference on a mode whose state is retained. Responses at two durations identify its source rate and reading gain; the inferred values choose a new time scale and force on the field already changed by calibration. The same finite certificate allows unknown centered read noise with a known variance cap. This is an additional response-based route, rather than a use of the independent-preparation variance laws on correlated readings.

7.1 Block scales as references for one another

Return to the three preparations of Table 1, denoted H_0, H_μ, and H_γ. One reading now means one scalar block mean from a freshly prepared field, rather than a complete coarse-field snapshot. Translation invariance gives its variance

$$\sigma_\ell^2(H) = \frac{1}{N_\ell} \sum_Q v_{H,\ell}(Q).$$

With a common unknown gain $b > 0$, the record has law $Y \sim N(0, b^2\sigma_\ell^2(H))$. All preparations are independent; the mean is known to be zero. Fix the reading counts $n_\ell, n_m > 0$ in advance and let S_ℓ, S_m be the sums of squares.

Proposition 7.2 (Gain-invariant source decisions)**.** *For every $b > 0$,*

$$R = \frac{S_\ell/n_\ell}{S_m/n_m} \overset{d}{=} \rho_H F_{n_\ell, n_m}, \qquad \rho_H = \frac{\sigma_\ell^2(H)}{\sigma_m^2(H)}. \tag{16}$$

Every function of the two positive sufficient statistics that is invariant under common positive rescaling depends only on R. With one such statistic, every invariant function is constant. Distinct ρ_H identify finitely many sources consistently as both reading counts tend to infinity, without a prior on b.

Proof. The variables $S_\ell/[b^2\sigma_\ell^2(H)]$ and $S_m/[b^2\sigma_m^2(H)]$ are independent chi-squares. Dividing each by its degrees of freedom gives the standard F law [5]. Positive rescaling is transitive on the positive reals; on positive pairs its orbits are labeled by the ratio. Finally, the two sample second moments converge to their population values, so $R \to \rho_H$ almost surely. □

For equal source weights, choose the largest scaled F density. This rule minimizes the average error among gain-invariant rules based on these sufficient statistics. When $n_\ell = n_m = n$, adjacent ordered ratios $\rho_i < \rho_{i+1}$ have decision boundary $\sqrt{\rho_i\rho_{i+1}}$: the scaled density is proportional to $\rho^{n/2}R^{n/2-1}/(\rho+R)^n$. Thus ordinary F cumulative probabilities give each source's exact finite-sample success probability.

At scales $(1, 32)$ the ratios for (H_0, H_μ, H_γ) are 12.85474928, 14.55261454, and 11.92780351. With 20,000 readings split equally, the success probabilities are .968359, .999036, and .969323. Table 2 gives the smallest even total for which all three reach .95 under the equal-split rule. These are numerical evaluations of the exact decision law, not Monte Carlo estimates or minima over all allocations.

Scale pair	Total independent readings
$(1, 32)$	15,984
$(1, 16)$	20,082
$(1, 8)$	36,732
$(1, 4)$	107,746
$(2, 32)$	96,336
$(4, 32)$	758,974

Table 2: Equal-split reading counts for at least .95 success under each source. Each scalar record requires a new field preparation; gain is fixed but unknown, and readout is noiseless.

The ratio removes gain from the decision; it does not make gain unidentifiable from the full data. Once the source is identified, its absolute variance determines b. Nor is R an unbiased estimate of ρ_H: its mean is $\rho_H n_m/(n_m - 2)$ when $n_m > 2$. The decision uses the distribution, not that mean. Preparation, block support, record precision, and subsequent storage remain separate resources. The fixed-count law cannot be reused unchanged after data-dependent stopping or allocation.

7.2 Readout noise and the price of a reference

A noisy probe changes both the cost and the distribution of a record. Consider independent Gaussian readout noise $\xi \sim N(0, r)$, with known variance. Two possible placements

give

$$Y = b(Z + \xi): \quad V = b^2(\sigma_\ell^2(H) + r), \tag{17}$$

$$Y = bZ + \xi: \quad V = b^2\sigma_\ell^2(H) + r. \tag{18}$$

These are alternative reading models. Take $r(E) = \nu/E$ with $\nu > 0$, and charge $E > 0$ per reading. This energy–noise relation is a specified probe model; preparation, memory, switching and the rest of the apparatus are not included in this charge.

Proposition 7.3 (Two kinds of calibration reference)**.** *For fixed settings in* (17)*, the gain-invariant ratio law holds with* $\rho_H = [\sigma_\ell^2(H) + r_\ell]/[\sigma_m^2(H) + r_m]$*. These ratios must be distinct to identify the sources by this rule. Moreover, at a single spatial scale, two known noise variances* $r_1 \neq r_2$ *identify both the gain and the source variance:*

$$b^2 = \frac{V_1 - V_2}{r_1 - r_2}, \qquad \sigma_\ell^2(H) = V_1/b^2 - r_1. \tag{19}$$

This identifies the source whenever its variances at that scale differ. The ratio law also applies to the two settings at one scale: with $\ell = m$ *and* $r_1 \neq r_2$*,* $\rho_H = [\sigma_\ell^2(H) + r_1]/[\sigma_\ell^2(H) + r_2]$ *is strictly monotone in* $\sigma_\ell^2(H)$*, so this reference also gives an exact, gain-free finite-sample decision.*

For fixed settings in (18)*, the raw sample ratio has law* $(V_\ell/V_m)F_{n_\ell,n_m}$*, generally dependent on* b*. Nevertheless, for every fixed* $b > 0$*,*

$$\frac{V_\ell - r_\ell}{V_m - r_m} = \frac{\sigma_\ell^2(H)}{\sigma_m^2(H)}. \tag{20}$$

Distinct source ratios therefore still identify the source and then the gain. Changing only the noise variance at a single scale in this second model identifies $b^2\sigma_\ell^2(H)$*, not its two factors.*

Proof. The chi-square argument applies to either pair of variances, including two settings at the same scale; there the derivative of $(\sigma^2+r_1)/(\sigma^2+r_2)$ in σ^2 is $(r_2-r_1)/(\sigma^2+r_2)^2 \neq 0$. In (17), subtracting the population variances at the two settings gives (19); in (18), subtracting the known noise gives (20). At a fixed scale in the second model, every setting has variance $c + r$ with the same unknown product c. All sources of positive variance can realize that product by changing b. □

The subtractions in (20) are statements about population variances. Finite sample estimates can fall below the noise floor and their subtracted ratio is not F distributed. Consistency at any fixed $b > 0$ follows from convergence of sample second moments; uniform accuracy down to arbitrarily small gains requires a separate resource analysis. A proper prior on gain can also give a single setting source information through absolute amplitude, despite the unknown-fixed-gain ambiguity.

Proposition 7.4 (Low-gain limit at fixed probe budget)**.** *Use* (18) *with* $r(E) = \nu/E$ *and two equally weighted sources* H, H'*. Fix the same* $b > 0$ *under both sources and a finite set of allowed scales. At each of at most* $T < \infty$ *readings, a policy* π *may choose the scale and energy using previous records and source-independent randomization, with* $\sum_i E_i \leq B$ *on every history. Each reading uses an independent new preparation, and*

no other source-dependent data are supplied. Put $\delta = \max_\ell |\sigma_\ell^2(H) - \sigma_\ell^2(H')|$. *For the resulting transcript laws,*

$$D_{\mathrm{KL}}(P_H^\pi \| P_{H'}^\pi) \le \frac{b^4\delta^2 B^2}{4\nu^2}, \qquad P_{\mathrm{err}}^\pi \ge \max\left\{0, \frac{1}{2} - \frac{b^2\delta B}{4\sqrt{2}\,\nu}\right\}. \tag{21}$$

The error bound applies to every terminal decision, including decisions based on adaptively chosen reads.

Proof. For centered Gaussian variances v, w,

$$D_{\mathrm{KL}}(N(0,v)\|N(0,w)) = \tfrac{1}{2}[v/w - 1 - \log(v/w)] \le \frac{(v-w)^2}{4\min(v,w)^2}.$$

This follows by Taylor's theorem for $-\log x$ between 1 and v/w. Conditional on a common transcript history, the policy uses the same distribution of settings under either source. Each conditional divergence is at most $b^4\delta^2 E_i^2/(4\nu^2)$, because $\min(v,w) \ge \nu/E_i$. The divergence chain rule and $\sum_i E_i^2 \le B^2$ give the first bound. Stopping can be padded with source-independent null records up to T. The equal-prior testing error is at least $(1 - \|P_H^\pi - P_{H'}^\pi\|_{\mathrm{TV}})/2$; Pinsker's inequality gives the second bound. □

The bound uses only $\sum_i E_i^2 \le B^2$, which is tight when the budget is spent on very few readings. If every reading also obeys $E_i \le E_{\max}$, then $\sum_i E_i^2 \le BE_{\max}$ and B^2 in (21) may be replaced by $BE_{\max}$; keeping the pathwise sum $\sum_i E_i^2$ gives a sharper statement still.

No allocation of this fixed probe budget, adaptive or otherwise, maintains reliable discrimination uniformly as $b \to 0$. The obstruction belongs to the placement (18). With the noise before the gain, (17), the same argument gives $|v - w| \le b^2\delta$ and $\min(v,w) \ge b^2\nu/E_i$, so the factor b^4 cancels. Fixed designs with gain-invariant decisions, and policies whose choices and decisions use only scale-invariant statistics, then perform identically at every gain; a policy that reacts to absolute amplitudes can still behave differently at different gains. This finite-resource obstruction coexists with identifiability at each fixed positive gain. Conversely, (19) shows that a controlled change in the probe can itself provide a reference. Cross-scale access is one way to resolve source–reader ambiguity; a known change in the reading interaction is another.

8 Self-reading at a chosen scale

Prediction and passive self-identification have the same statistical quantity when the competing interface produces independent records with the same one-record law. For a stationary record law P and that independent law $Q^{\otimes n}$, the divergence chain rule gives

$$D(P^n\|Q^{\otimes n}) = nD(P_1\|Q) + \sum_{j=2}^{n} I_P(Y_j; Y_1, \ldots, Y_{j-1}). \tag{22}$$

When $P_1 = Q$, the divergence per record tends to $I_P(Y_0; Y_{-\infty}^{-1})$, possibly infinite. This is the passive test in *The World as Imprint* [6], not an upper bound on all active identification protocols. It also gives Theorem 6.1 a self-reading interpretation: the increase from one past Gaussian record to two is $\frac{1}{2}\log(R_1/R_2) > 0$ exactly when the alias rates differ.

Now sample one physical block mean along a stationary trajectory of (6), at interval Δ, and add independent Gaussian readout noise of variance $r \geq 0$. Write

$$w_\ell(k) = \frac{|W_\ell(k_1)W_\ell(k_2)|^2}{L^2\ell^4 K(k)}, \qquad a_k = e^{-K(k)\Delta}, \qquad v_\ell = \sum_k w_\ell(k).$$

The record spectral density is

$$f_{\ell,r}(\omega) = r + \sum_k w_\ell(k) \frac{1 - a_k^2}{1 - 2a_k \cos\omega + a_k^2}. \tag{23}$$

The positive finite mixture has absolutely summable correlations and a strictly positive smooth density. The Gaussian entropy-rate formula therefore applies [3].

Proposition 8.1 (The reader's noise changes its informative scale)**.** *Against independent Gaussian records of variance $v_\ell + r$, the passive divergence rate, in nats per continuous record, is*

$$\mathcal{I}_\ell(r) = \frac{1}{2}\log(v_\ell + r) - \frac{1}{4\pi}\int_{-\pi}^{\pi} \log f_{\ell,r}(\omega)\, d\omega. \tag{24}$$

It is nonincreasing in r at each scale. The scale maximizing it need not remain fixed when the reader's noise changes.

Proof. The independent law has entropy rate $\frac{1}{2}\log[2\pi e(v_\ell + r)]$, and its cross-entropy under the matched law is the same because their one-record marginals agree. Subtract the matched Gaussian entropy rate to get (24). Alternatively this is the limiting predictive information in (22). Adding independent noise of variance $r_2 - r_1$ to every record maps both candidate laws at r_1 to those at r_2. Data processing for each finite transcript and passage to the rate prove monotonicity. The table below supplies the change of maximizing scale. □

r	$\ell = 1$	2	4	8	16	32	64
0	.3405	.6219	.9486	1.2351	1.4254	1.5250	1.6194
10^{-4}	.3404	.6210	.9449	1.2200	1.3684	1.3390	1.1516
10^{-2}	.3229	.5479	.7092	.6769	.4599	.2076	.0688

Table 3: Passive self-reading rate for $L = 64$, $\mu = .1$, $g = \gamma = 0$, $\Delta = .2$. Each row compares fixed ideal continuous-record interfaces, not a finite-memory controller or a total-cost optimum.

The best listed scale moves from 64 to 16 to 4. Coarse means have longer temporal correlations, but smaller variance; fixed readout noise masks them first. After noise rises from 10^{-4} to 10^{-2}, staying at scale 16 gives .4599 nats per record, whereas scale 4 gives .7092. These numbers assume known noise. They do not by themselves identify the damage, choose a scale online, or give the support and apparatus bill for the new interface.

9 A retained history chooses the next reading scale

We next make the allocation depend on records. This experiment uses fresh preparations, not the stationary trajectory of the preceding section. At $L = 64$ set $H_0 = (\mu, \gamma) = (.1, 0)$ with $g = 0$. Port 0 reads the global mean, of variance $1/(L^2\mu)$. Port 1 reads one site, of variance $v(\mu, \gamma) = L^{-2} \sum_k K_{\mu,\gamma}(k)^{-1}$. Let

$$H_0^{\mathrm{alt}} = (.09, \gamma_\mu), \qquad v(.09, \gamma_\mu) = v(.1, 0), \qquad H_1^{\mathrm{alt}} = (.1, 1/12).$$

The positive solution is $\gamma_\mu \simeq .0144867958104$; monotonicity in γ makes it unique. The equality, rather than this rounded number, defines the preparation. Port 1 is exactly blind to H_0^{alt}, and port 0 to H_1^{alt}.

In episode t, a hidden regime $J_t \in \{0, 1\}$ selects the available alternative. The source is H_0 or $H_{J_t}^{\mathrm{alt}}$ with equal probability, independently of earlier source flags given the regime. The regime starts uniform and changes with probability $0 < \varepsilon \le 1/2$ between episodes. Each reading uses a fresh field and returns one bit: $|Y| > \tau_b \sqrt{v_b(H_0)}$ at port b, with $\tau_0 = 1.5222951844903065$ and $\tau_1 = 1.445716072545341$. The baseline and informative-alternative probabilities are

$$(p_0^0, p_0^1) \simeq (.1279351341, .1486894695), \qquad (p_1^0, p_1^1) \simeq (.1482568544, .1287055577). \quad (25)$$

At the blind port the alternative probability is exactly the baseline. The thresholds are specified designs, not claimed optimal quantizers. All policies have $n = 4000$ readings and preparations per episode and choose the episode's split before collecting its records. The task is to report whether that episode's source is the alternative. Equal read counts do not equate the spatial support of a global mean and a site measurement. This comparison stipulates equal read and preparation prices; physical support and full apparatus energy remain separate resources.

Theorem 9.1 (Fixed-allocation bound and history-dependent allocation)**.** *Define the one-bit Bhattacharyya exponents*

$$\beta_b = -\log\left(\sqrt{p_b^0 p_b^1} + \sqrt{(1 - p_b^0)(1 - p_b^1)}\right), \qquad \beta_{\max} = \max_b \beta_b.$$

1. *Every fixed split, or mixture of fixed splits independent of the regime, has average error at least*
$$B_n = \frac{1}{2}\left(1 - \sqrt{1 - e^{-n\beta_{\max}}}\right), \quad (26)$$
even if the decoder is told the current regime and keeps all past records. For the stated experiment $B_{4000} \simeq .042754$.

2. *A history policy with five retained bits has a finite Markov description and a computable stationary error. One explicit policy at $\varepsilon = 10^{-4}$ has stationary error numerically evaluated as .034233, below (26). Thus its advantage is over every fixed split, not only a tested set of allocations.*

3. *At $\varepsilon = 1/2$, history-dependent episode allocations cannot beat the best history-free allocation under the same episode protocol. This statement does not restrict policies which reallocate after readings within the episode.*

Proof. For m readings at an informative port, the two product laws have Bhattacharyya coefficient $e^{-m\beta_b}$. Cauchy–Schwarz bounds their total variation by $\sqrt{1-e^{-2m\beta_b}}$, so the equal-prior Bayes error is at least $b(m) = \frac{1}{2}(1-\sqrt{1-e^{-2m\beta_{\max}}})$. With the regime given, the other port carries no source information; a split $(m, n-m)$ therefore has error at least $[b(m)+b(n-m)]/2$. The decreasing function b is convex on $(0,\infty)$ and continuous at zero. Jensen's inequality gives $b(n/2) = B_n$; independent randomization preserves the bound. Past records cannot reveal the fresh source flag once J_t is given.

For part (2), retain a believed regime $b \in \{0,1\}$ and a counter $c \in \{1,\dots,16\}$. Read only b. A coarse count at least 553 or a fine count at most 553 declares the alternative. A detection raises c, capped at 16. Without a detection, an independent fair coin lowers c with probability $1/2$; at zero switch b and set $c = 1$. Nothing in this rule uses J_t or the source flag. Its transition matrix on (J, b, c) has 64 states. A binomial tail gives each detection probability; the source flag is averaged with weight $1/2$ and the regime changes after the update. Solving $\pi T = \pi$, $\sum \pi = 1$, and averaging the conditional task error gives the stated numerical value. This is a deterministic evaluation of a specified finite chain, not an adaptive optimality proof.

For part (3), J_t is independent of all previous records and private coins. Conditional on the history the allocation is fixed, and current source data have their history-free law. The conditional Bayes error cannot be below its minimum over fixed allocations. Averaging proves the claim. □

The sharper minimum with the regime given is obtained by evaluating $[e_0(m) + e_1(4000-m)]/2$ for all 4001 splits, where

$$e_b(m) = \frac{1}{2}\sum_{k=0}^{m} \min\{\mathrm{Bin}_{m,p_b^0}(k), \mathrm{Bin}_{m,p_b^1}(k)\}.$$

It is .095897 at $m = 1965$. The analytic bound, although less sharp, already separates the history policy from the whole fixed class. The same counter rule, with explicitly chosen cap C, yields:

ε	C	Stationary error
10^{-5}	16	.032754
10^{-4}	16	.034233
10^{-3}	8	.041516
10^{-2}	4	.081334

These are attainable policies, not minima over all histories or universal switching thresholds. The unknown common gain problem of Section 7.1 has not been added to this experiment.

Corollary 9.2 (A price interval for retained history)**.** *Give each correct source report value $V > 0$ and define the exact finite-comparison value*

$$E_{\mathrm{fix}} = \min_{0\le m\le n} \frac{e_0(m) + e_1(n-m)}{2}.$$

If the history policy's additional memory, randomization and update charge per episode obeys

$$0 \le \lambda < V(E_{\mathrm{fix}} - L_H),$$

where L_H is its stationary error, its net value exceeds that of every fixed allocation, including mixtures, even with a regime-informed decoder. The analytic bound $E_{\mathrm{fix}} \geq B_n$ gives the closed-form subinterval $0 \leq \lambda < V(B_n - L_H)$. At $\varepsilon = 10^{-4}$, evaluation of the exact finite formulas gives $E_{\mathrm{fix}} = .0958969785$, $L_H = .0342328069$ and $(E_{\mathrm{fix}} - L_H)V \simeq .06166417V$, compared with the analytic subinterval of approximately $.008521V$. Displayed rounding is not an interval certificate.

Proof. At each fixed split the best regime-informed decoder has error $[e_0(m) + e_1(n - m)]/2$: its other port is blind and the fresh source flag is independent of past records conditional on the current regime. A mixture of fixed splits cannot lower their minimum. Subtract the common preparation and read charges; the gain in correct reports is at least $V(E_{\mathrm{fix}} - L_H)$. Charging λ gives the result. □

The public supplement `supplemental/history_price` contains the finite-state calculation and fixed-allocation enumeration for this corollary. Here λ is a task-value charge, not a thermodynamic estimate. The retained automaton has 32 states and needs five bits. Its within-episode count has 4001 states and needs twelve more transient bits; acquisition timing, model storage, randomization, resetting and computation remain in the apparatus bill. Silence is informative because the contrast occurrence rate is specified: repeated absence of an expected detection changes the future port choice. A rule which switches only on a detection at the unread port never switches when it takes no probe there. Such a frozen belief has long-run error $\frac{1}{2}e_b(4000) + \frac{1}{4}$, exceeding .264. This is a failure of that update rule, not a claim that all zero-probe policies fail. The constructed policy uses silence and needs no second-port readings in these slowly changing regimes.

10 A history chooses how to read and act on the same field

The preceding allocation experiment scores a report. Here a reading is followed by an operation on its physical source. Take the stationary process (6), with $K > 0$ the finite real-space precision matrix. Retain $R = a^T X_{-\Delta} + \xi_R$, where a is a block vector. At time zero either acquire one $Z_\ell = h_\ell^T X_0 + \xi_\ell$ or take no new reading. Independent Gaussian read noises have known positive variances. These classical readings are assumed not to disturb X; their apparatus and dissipation are not derived from the noise law.

After that decision, apply a constant force Bu for $0 < s < \Delta$, with $B = a/\|a\|^2$ and $u \in \{-U, 0, U\}$. Then

$$dX_s = (-KX_s + Bu)\,ds + \sqrt{2}\,dW_s, \qquad Y_\Delta^u = Y_\Delta^0 + \alpha u, \qquad \alpha = a^T K^{-1}(I - e^{-K\Delta})B > 0, \tag{27}$$

where $Y = a^T X$. The zero superscript is the unforced continuation of the same field and bath noise, not a new preparation. In units with bath temperature and mobility one, the force comes from the nonnegative potential $H_u(X) = \frac{1}{2}(X - K^{-1}Bu)^T K(X - K^{-1}Bu)$. The force pattern, finite amplitude, clock and read ports are inputs. The task minimizes $\mathbb{E}[(Y_\Delta^u)^2 + c_\ell]$, with zero read charge when no read is acquired. These c_ℓ are task prices, not heat.

Theorem 10.1 (Reading for a bounded physical action)**.** *Suppose that, conditional on the authorized history $\mathcal{H}$, the unforced target and candidate readings are jointly Gaussian.*

Let $m = \mathbb{E}[Y_\Delta^0 \mid \mathcal{H}]$ *and put*

$$q_\ell = \frac{\operatorname{Cov}(Y_\Delta^0, Z_\ell \mid \mathcal{H})^2}{\operatorname{Var}(Z_\ell \mid \mathcal{H})}, \quad D = \alpha U, \quad \mathcal{B}(m) = 2D(|m| - D/2)_+.$$

For $N \sim N(0,1)$ *define*

$$G_\ell(m) = \mathbb{E}\mathcal{B}(m + \sqrt{q_\ell}N) - \mathcal{B}(m) - c_\ell. \tag{28}$$

Choose the reading with the largest positive G_ℓ*, or no reading if none is positive. Let* M *be the revised conditional mean of the unforced target. Apply zero force for* $|M| \le D/2$*, and* $u = -U \operatorname{sign} M$ *otherwise. The risk is*

$$\mathbb{E}[(Y_\Delta^0)^2] - \mathbb{E}\mathcal{B}(m) - \mathbb{E}\max\{0, G_1(m), \dots, G_p(m)\}. \tag{29}$$

This minimizes the one-round objective over every measurable choice using that history, at most one permitted new reading and the stated three forces.

Proof. For mean M, the force changes conditional square loss by $2\alpha u M + \alpha^2 u^2$. Its minimum is $-\mathcal{B}(M)$. Gaussian conditioning makes the revised mean, before receiving the new result, distributed as $m + \sqrt{q_\ell}N$. The unforced target's conditional second moment averages to its previous value. Thus the reading's reduction is exactly G_ℓ. Maximize pointwise and average to obtain (29). □

Convexity of $\mathcal{B}$ makes the uncharged reading benefit nonnegative. Its priced value depends on the history and on the changes of optimal action at $M = \pm D/2$, not just on an information rate. This is Bayesian experimental decision-making on a scale-filtered carrier. Informative control is an established subject [1]; the present Gaussian law and finite force menu make its composition with imprint reading explicit.

10.1 A finite-record case

Set $L = 32$, $\mu = .1$, $g = \gamma = 0$, and $\Delta = .5$. The target and old reading use a 4×4 block. Current ports use nested 2×2 and 4×4 means. Every read noise variance is .001. Choose $D = .15$, $c_2 = .001$, $c_4 = .014$. Finite lattice sums give

$$v = \operatorname{Var}(Y) = .1443687793, \qquad m = .6983747950R,$$
$$\alpha = .4012307531, \qquad U = .3738497083.$$

and $q_2 = .0079283594$, $q_4 = .0272698232$. The reader chooses scale 4 for $|R| < .123905791$, scale 2 for $.123905791 < |R| < .285137090$, and no read outside. Equalities may be assigned either way. The thresholds are roots of the Gaussian value functions, not fitted rules. With $d = D/2$,

$$\mathbb{E}\mathcal{B}(m + \tau N) = 2D \sum_{\epsilon = \pm 1} \left[\tau\phi\left(\frac{\epsilon m - d}{\tau}\right) + (\epsilon m - d)\Phi\left(\frac{\epsilon m - d}{\tau}\right) \right]. \tag{30}$$

Every choice independent of R is a mixture of the first three rows, even if its force decoder later uses R and the new result. It cannot beat their minimum. The history-selected rule does. Its material square loss is .093688 and its mean read charge .003858. If all information from R is removed before the choice and action, the best risk in this same menu is .103835. Neither comparison allows an extra read or a field reset.

The strict advantage survives sixteen-bit records, finite coefficient and threshold precision, and displacement error at most 10^{-5}. Appendix B gives the implementation and register account, certifying risk below .097763 against the ideal fixed-choice minimum .098280. The analog interface, clock and physical storage remain additional resources.

Next reading	Mean reads	Task risk	Switching work
None	0	.100621	.91202
Always scale 2	1	.098280	.94975
Always scale 4	1	.103732	1.02893
History-selected	.54545	.097546	1.02916

Table 4: One continuation of the same field. Every row retains the old reading and uses its optimal permitted force. Task risk includes the read price; switching work is in bath-energy units and excludes the reader and memory apparatus.

10.2 Work and continued operation

Switch H_u on after reading and off at Δ. The force-dependent constant cancels between switches, giving the pathwise mechanical work

$$W_{\mathrm{sw}} = uB^T(X_\Delta^u - X_0). \tag{31}$$

This is the rectangular switching protocol in Table 4, not c_ℓ. Finite ramps need not destroy the strict comparison. Replace the force by $Buf(s)$, with f rising linearly from zero to one in time τ, remaining one and falling to zero in the final τ. The response becomes $\alpha_f = a^T \int_0^\Delta e^{-K(\Delta-s)} Bf(s)\,ds$. Since $K > 0$ and $\|a\|\|B\| = 1$, $|\alpha - \alpha_f| \le 2\tau$. Choosing $\tau = 5 \times 10^{-6}/(2U)$ bounds the displacement change by 5×10^{-6}; the remaining 5×10^{-6} may cover calibrated amplitude error. This fits the finite-controller certificate. The ramps have finite slope, though their timing and drive apparatus remain supplied. Their actual work is $-u \int_0^\Delta f'(s) B^T X_s\,ds$, obtained from the same potential; the force-dependent offset integrates to zero. The history-selected rule improves the declared objective but uses more switching work than always reading scale 2. It is not a total-energy optimum. The measurement and memory parts of an information-thermodynamic account require their own physical implementation [2].

The resulting field can be read again. Known K, linear noisy readings and retained controls give a Gaussian conditional-state recursion on the same carrier [4]. After each force its mean and covariance must be propagated; one cannot reapply the old stationary prior or thresholds without that update. Reusing the one-round rule is not a long-horizon optimality proof. Likewise, the finite-record certificate above concerns this single continuation, not an indefinitely running sixteen-bit controller.

11 A control becomes a calibration reference on the same carrier

The unknown-gain calculations used independent preparations. A control can instead supply a reference on the field already being read, without replacing its changed state by equilibrium. Choose one normalized real spectral coordinate of the lattice, with known Laplacian eigenvalue s_0, set $g = \gamma = 0$, and let its unknown stable relaxation rate be $\lambda = \mu + s_0 > 0$. A known force profile has projection $B > 0$ on this mode. On consecutive intervals it obeys

$$dX_t = (-\lambda X_t + Bu_i)\,dt + \sqrt{2}\,dW_t, \qquad Z_i = bX_{t_{i+1}} + \xi_i,$$

where $b > 0$ is unknown and fixed, ξ_i is independent centered Gaussian read noise of known variance r. The mode, B, clock and gain are the same for both reading durations. A force pattern or gain allowed to change with the duration would define another problem.

Before each interval, use a fresh fair coin to choose $u_i = \pm U$, $U > 0$, independently of the entire past carrier, hidden parameters and retained history. Alternate durations Δ and 2Δ, with $\Delta > 0$; neither the field nor the bank of old records is reset. Put $f_\tau(\lambda) = (1 - e^{-\lambda\tau})/\lambda$.

Proposition 11.1 (Source and reader identified by paid responses)**.** *For an interval of duration τ, conditional on its full physical past,*

$$\mathbb{E}[u_i Z_i/U^2 \mid \textit{past}] = \kappa_\tau = bBf_\tau(\lambda).$$

Thus the two response moments identify the relaxation rate and gain:

$$q = \frac{\kappa_{2\Delta}}{\kappa_\Delta} - 1 = e^{-\lambda\Delta}, \qquad \lambda = -\frac{\log q}{\Delta}, \quad \mu = \lambda - s_0, \quad b = \frac{\kappa_\Delta}{Bf_\Delta(\lambda)}. \tag{32}$$

With known signed B, the response sign would also identify a nonzero signed gain, unlike centered variance data. The finite certificate below uses a positive gain interval. If B is unknown, only the product bB is identified by these responses.

For a uniform finite-sample certificate, suppose $\lambda \in [\lambda_-, \lambda_+]$, $b \in [b_-, b_+]$ with $\lambda_-, b_- > 0$, and $\mathbb{E}X_0^2 \le V_0 < \infty$. Define

$$V_* = \max\left\{V_0, \frac{1}{\lambda_-} + \left(\frac{BU}{\lambda_-}\right)^2\right\}, \qquad Q = \frac{b_+^2 V_* + b_+^2/\lambda_- + r}{U^2}.$$

Collect n intervals of each duration and average $u_i Z_i/U^2$ separately into $\widehat{\kappa}_\Delta, \widehat{\kappa}_{2\Delta}$. For any $\eta > 0$, both moment errors are at most η except on an event of probability at most $\min(1, 2Q/(n\eta^2))$. Let $\kappa_- = b_- Bf_\Delta(\lambda_+)$ and assume $\eta < \kappa_-$. Clip the estimated q to $[e^{-\lambda_+\Delta}, e^{-\lambda_-\Delta}]$ before taking its logarithm. On the certified event,

$$|\widehat{\lambda} - \lambda| \le e_\lambda := \frac{3\eta}{(\kappa_- - \eta)\Delta e^{-\lambda_+\Delta}}, \tag{33}$$

$$|\widehat{b} - b| \le e_b := \frac{\eta}{Bf_\Delta(\lambda_+)} + \frac{b_+\Delta^2}{2f_\Delta(\lambda_+)} e_\lambda. \tag{34}$$

Clipping $\widehat{b}$ to $[b_-, b_+]$ preserves the latter bound.

Proof. The exact interval solution is $X_{t+\tau} = e^{-\lambda\tau}X_t + Bf_\tau(\lambda)u_i + \zeta_i$, with independent centered innovation of variance $(1 - e^{-2\lambda\tau})/\lambda$. The fresh coin cancels the old-state term, even when that state is nonzero or correlated with all old records. Multiplication by u_i and conditional expectation give the response identity; the ratio $f_{2\Delta}/f_\Delta = 1 + e^{-\lambda\Delta}$ proves identification.

The second-moment recursion is

$$\mathbb{E}X_{t+\tau}^2 = e^{-2\lambda\tau}\mathbb{E}X_t^2 + B^2U^2 f_\tau(\lambda)^2 + (1 - e^{-2\lambda\tau})/\lambda.$$

The chosen V_* is invariant for either duration, since $f_\tau(\lambda)^2/(1-e^{-2\lambda\tau}) = (1-e^{-\lambda\tau})/[\lambda^2(1+e^{-\lambda\tau})] \le 1/\lambda_-^2$. The centered response samples are martingale differences with conditional variance

$$U^{-2}\left[b^2 e^{-2\lambda\tau}X_t^2 + b^2(1 - e^{-2\lambda\tau})/\lambda + r\right].$$

Their unconditional second moments are at most Q. Orthogonality gives each average variance at most Q/n, despite dependence of the carrier across intervals. Chebyshev's inequality and a union bound give the stated probability. The true response ratio lies in $(1, 2)$, so its error is at most $3\eta/(\kappa_- - \eta)$. Clipping q cannot increase error, and log is $e^{\lambda_+\Delta}$-Lipschitz on that interval. Finally $|f'_\Delta| \le \Delta^2/2$ and $f_\Delta \ge f_\Delta(\lambda_+)$ give (34). □

Corollary 11.2 (Response calibration with unknown read noise)**.** *In Proposition 11.1, replace the Gaussian read errors by mutually independent centered errors, independent of the carrier past, bath increments and force coins, with $\mathbb{E}\xi_i^2 \le r_+ < \infty$. Their distributions and actual variances may be unknown and may differ between readings. The response identity, source–gain identification and finite parameter-error certificate remain valid with r_+ in place of r in Q.*

Proof. The read error contributes zero to the conditional response mean. Independence removes its cross terms with the old-state and bath terms. Its contribution to the conditional variance is $\mathbb{E}\xi_i^2/U^2 \le r_+/U^2$. The carrier's second-moment recursion is unchanged because read errors do not enter its force coin or dynamics. The centered responses are therefore still martingale differences with the same upper bound Q after the substitution. Orthogonality, Chebyshev's inequality and the deterministic ratio and logarithm bounds prove the claim. □

This extension does not identify the read-noise law. Nor does it extend the Gaussian likelihoods, entropy rates or conditional-state filters used elsewhere in the paper. With unknown parameters, or non-Gaussian read noise, those Gaussian state updates cannot be recovered by inserting fitted parameters and a variance cap. The response added by the next force, however, depends on the same linear dynamics and conditional zero mean, so Corollary 11.4 also uses this certificate. Without a variance cap the population response identity can still hold, but the uniform finite-budget bound above is unavailable.

Corollary 11.3 (Finite records for response calibration)**.** *In Corollary 11.2, clip each endpoint record to $[-A, A]$ and round to $2^{B_{\mathrm{rec}}}$ equally spaced levels, with step $\delta = 2A/(2^{B_{\mathrm{rec}}} - 1)$. Put $M_Z = b_+^2 V_* + r_+$ and $e_{\mathrm{rec}} = [\delta/2 + M_Z/(4A)]/U$. For any $\eta, \epsilon > 0$ with $\eta + \epsilon < \kappa_-$, the parameter and next-action bounds remain valid with $\eta + \epsilon$ in place of η, except on an event of probability at most*

$$\min\left\{1, \frac{2Q}{n\eta^2} + \frac{2e_{\mathrm{rec}}}{\epsilon}\right\}.$$

Thus a sufficiently large but finite range and record register retain every fixed positive-tolerance certificate.

The proof is in Appendix A.1. The accumulation can be exact on integer level indices: a signed sum for each duration needs at most $B_{\mathrm{rec}} + \lceil \log_2(n+1) \rceil + 2$ bits, with the record range, force scale and subsequent parameter arithmetic supplied separately. Quantization does not manufacture an analog measuring device or a cost-free cyclic processor.

No independent-read variance formula was used. The fresh randomness creates orthogonal response errors, not independent field samples. The price is $2n$ actual force intervals and endpoint readings, $3n\Delta$ elapsed time, the specified spatial force/read support, coins, moment accumulation and retained records. Corollary 11.3 gives the additional finite-record error and the accumulation storage needed for this certificate. For each fixed positive gain one can use its actual finite constants. Without a positive lower gain bound

this is not a uniform fixed-budget claim as $b \to 0$. For failure probability at most δ_{fail}, the sufficient count is $n \geq 2Q/(\delta_{\text{fail}}\eta^2)$. This Chebyshev budget is conservative, not an efficiency or optimal-design claim.

Corollary 11.4 (Calibration selects the next time scale and action)**.** *Permit force amplitudes in* $[0, U_{\max}]$*, rather than only the three-action menu of Section 10. Let* $q_* \in (0,1)$ *be a desired response decay. After the calibration history, set* $\widehat{\tau} = -\log(q_*)/\widehat{\lambda}$*. On the preceding certified event,*

$$|\widehat{\tau} - \tau_*| \leq \frac{-\log(q_*)}{\lambda_-^2} e_\lambda, \qquad \tau_* = -\log(q_*)/\lambda.$$

For a desired readout displacement $0 \leq D_* \leq U_{\max} b_- B(1-q_*)/\lambda_+$*, install the legal force*

$$\widehat{u} = \frac{D_* \widehat{\lambda}}{\widehat{b}B(1-q_*)} \leq U_{\max}$$

for this next interval. With $\tau_{\max} = -\log(q_*)/\lambda_-$*, its actual mean displacement relative to the unforced continuation of the same current state differs from* D_* *by at most*

$$U_{\max} B \left[\tau_{\max} e_b + b_+ \tau_{\max}^2 e_\lambda/2\right].$$

Proof. The reciprocal function is λ_-^{-2}-Lipschitz. The predicted displacement is exactly $\widehat{b}Bf_{\widehat{\tau}}(\widehat{\lambda})\widehat{u} = D_*$. At that same actual duration, $|bBf_{\widehat{\tau}}(\lambda) - \widehat{b}Bf_{\widehat{\tau}}(\widehat{\lambda})| \leq B[\tau_{\max} e_b + b_+\tau_{\max}^2 e_\lambda/2]$. Multiply by the legal amplitude. □

The comparison concerns the response added to a current, generally nonstationary field. It does not erase the calibration's perturbation. The old state contributes to the absolute next record and must be propagated. Once unknown parameters are inferred from these records, a posterior over parameters and state is generally a mixture, not one Gaussian state. The known-parameter Gaussian rule of Theorem 10.1 may not be applied to that mixture by silently replacing it with a fitted equilibrium law.

Force ramps also belong to the model. A normalized taper replaces f_τ by $\int_0^\tau e^{-\lambda(\tau-s)} f(s)\, ds$; arbitrary tapers do not preserve (32). The exact identity above uses the stated rectangular force. Its potential is nonnegative, $H_u(X) = \lambda(X - Bu/\lambda)^2/2$, and a switch from u to v costs the actual potential difference at the current X, not a price attached to an observation. Finite amplitudes and durations do not by themselves supply a heat, memory-reset or complete-apparatus work ledger.

12 From trace-carrying to organized reading

A source contrast, its carrier and the available reading are distinct. The zero mode can miss a contrast present in the field; one spectral band can confuse a source change with reader gain. Temporal filtering can instead make past records useful when the coarse present is insufficient. This supplies a precise retention-and-reuse interface to Stable Hulls [10], with a source-specific meaning of access dependence [11].

Records also change the next physical operation. The regime protocol selects a port, the Gaussian continuation selects a reading and force, and paid responses calibrate a future duration and command on a changed carrier. Their preparation and noise laws

remain different; unknown gain, reader drift and hidden switching cannot be combined by identifying their estimates. The Imprint Chain supplies separate use and maintenance permissions [4]; this field does not derive its body's repair law. Port formation, control access and first-person experience remain separate explanatory questions.

13 Conclusion

Scale-stable readability is not microscopic reconstruction. Block averaging in the specified Gaussian field retains substantial mass information while suppressing higher-gradient contrasts. The same operation leaves a dynamic consequence: past and future are conditionally correlated when the coarse present is insufficient. A finite retained record exploits that correlation, with strictly better risk than any current-only predictor. Additional spectral access separates a source parameter from the reader's own gain. Independent block readings supply an exact finite-sample calibration law. Noisy readout distinguishes two further questions: whether a source is identifiable at all, and whether the available probe budget can identify it reliably. Known probe variations can supply a calibration reference even at one spatial scale.

Filtering now has two consequences for a reader. It gives retained history a prediction value, and it changes which scale is worth reading when the source or reader changes. A finite history can govern the next allocation and beat every fixed allocation under an explicit price interval. On a single continuing field, it can also choose a noisy block reading and a bounded physical force. The optimal one-round rule, its finite-record advantage and its switching work make prediction, inquiry and actual change parts of the same experiment. The active response reference adds a further step: a history can learn which source and reader parameters govern its actual intervention, then select a new time scale and command on the changed carrier. The moment certificate prices a finite experiment but does not optimize it. The formation of ports, known force relations and maintenance capacities remains a physical question. Unknown centered independent read noise with a known variance cap preserves this response route, without turning the other Gaussian inference protocols into noise-independent laws.

A Numerical evaluation

Spectral sums use the complete finite grid with analytic derivatives. Chernoff optimization is one-dimensional on $[0, 1]$. For stability its per-mode summand is evaluated as $\frac{1}{2}\{\log[1 + u(e^z - 1)] - uz\}$, where $z = \log(v_1/v_0)$, using `log1p` and `expm1`. Finite bins are integrated before computing their discrete Chernoff coefficient.

For an 8×8 lattice, direct physical-space covariance and block matrices agree with (2) to 3.6×10^{-15}. A sample of 6000 fields at $L = 32, \ell = 4$ has mean measured-to-predicted spectral variance ratio .995762, without removing an overall normalization. Direct Gaussian regression agrees with (11) to about 10^{-17}.

For Table 2, scalar block variances are also evaluated directly on the fine lattice as

$$\sigma_\ell^2(H) = \frac{1}{L^2 \ell^4} \sum_k \frac{|W_\ell(k_1) W_\ell(k_2)|^2}{K_H(k)}.$$

The equal-count geometric decision boundaries give success probabilities from F cumulative and survival functions. Searching integer counts and checking each preceding count

locates the listed thresholds; no density-grid quadrature is used.

Table 3 uses the finite Fourier transform of each physical block and the spectral density (23). Periodic midpoint grids of 8192 and 16384 frequencies agree within 9×10^{-16} for these listed rates. This agreement is a numerical convergence check, not an interval certificate for arbitrary spectra. The history policies use analytic binomial tails, an enumerated finite transition matrix and its normalized stationary solution; the stationarity residual is below 3×10^{-16}. The fixed regime-informed comparison enumerates all integer splits. Its likelihood-threshold tails agree with direct binomial sums through $m = 32$ within 9×10^{-16}. No Monte Carlo estimate or policy search is needed for the stated constructed-policy comparisons.

For the active response reference, conditional means were checked by explicit averaging over the two forces at four nonzero or zero old-state values. A diagnostic uses 64 continuing carriers, each with 20,000 intervals at each duration, starting at $X_0 = 2$. At $(\lambda, b, B, U, \Delta, r) = (.7, 1.3, .8, .6, .5, .01)$, population response moments are .4387491 and .7479304; empirical replica means are .4416915 and .7465218. Their comparison uses the martingale variances accumulated from the actual second-moment recursion, not an independent-field approximation. Perturbed response moments at the four corners of a fixed error square satisfy the parameter and next-action bounds. A force sign aligned with the old state instead gives a nonzero response bias, confirming the independence requirement. These checks do not certify force apparatus heat or optimize the conservative budget.

A.1 Finite response records

Proof of Corollary 11.3. For the quantizer q, the pathwise error is bounded by $\delta/2+(|Z|-A)_+$. The inequality $(|z|-A)_+ \le z^2/(4A)$ and $\mathbb{E}Z_i^2 \le M_Z$ give $\mathbb{E}|\mathsf{q}(Z_i) - Z_i| \le \delta/2 + M_Z/(4A)$. Each duration's response average therefore has mean absolute quantization error at most e_{rec}. Markov's inequality and a union bound put both such errors below ϵ with failure at most $2e_{\mathrm{rec}}/\epsilon$. Combine this with the unquantized martingale event. The remaining ratio, clipping and action bounds are deterministic. Taking A large and then B_{rec} large makes e_{rec} arbitrarily small; no Gaussian tail law was used. □

B Finite records for the reading-and-force rule

B.1 Rational arithmetic for the prediction rule

An entirely finite-arithmetic variant uses rational $A = 3$, the same 10-bit records, and rounds a_1, a_2 and the output to the nearest 2^{-16}. Intermediate rational arithmetic is finite. Coefficient and output rounding add at most $(2A+1)2^{-17}$ to the predictor error. Adding that amount to $(|a_1| + |a_2|)\eta$ before squaring in (13) gives risk at most $.021433187 < R_1$. The parameters are known and the extra registers must be counted. The advantage therefore does not require an infinite-precision archive, though it remains relative to a specified model and interface.

B.2 The reading-and-force controller

Both old and newly acquired scalar records use the nearest-point uniform quantizer on $[-4, 4]$ with 2^{16} levels, including the endpoints, and saturation outside the interval. De-

coder coefficients are rounded to sixteen binary fractional bits; the two R-thresholds to twenty, giving $\epsilon_R \le 2^{-21}$. This is distinct from the ten-bit prediction variant above. There are sixteen retained data bits and sixteen transient data bits; five fixed decoder coefficients use eighty fractional bits in total and the two thresholds forty. A forty-bit signed accumulator and a twenty-one-bit revised-mean register suffice for this arithmetic construction. Program, acquisition clock, analog interface and physical storage remain additional resources; these are sufficient register sizes, not minimal ones.

Quantization and decision regret. For a centered Gaussian record of variance V, clipping and uniform quantization on $[-A, A]$ have L^1 error at most

$$\eta(V) = \frac{A}{2^B - 1} + \sqrt{2V[z\phi(z) + \overline{\Phi}(z)]}, \qquad z = A/\sqrt{V}.$$

Use the L^2 clipping bound of Corollary 6.2 and Cauchy–Schwarz. Write the old mean as βR and the revised mean as $b_\ell R + d_\ell Z_\ell$, where

$$d_\ell = \frac{\operatorname{Cov}(Y^0_\Delta, Z_\ell \mid R)}{\operatorname{Var}(Z_\ell \mid R)}, \qquad b_\ell = \beta - d_\ell \frac{\operatorname{Cov}(Z_\ell, R)}{\operatorname{Var} R}.$$

Include $b_0 = \beta$ for no reading, and set $C_R = \max_{\ell \ge 0} |b_\ell|$.

The function $\mathcal{B}$ is $2D$-Lipschitz, and each G_ℓ is $4D$-Lipschitz. Maximizing scores at a perturbed old mean costs at most twice their largest perturbation. Choosing the force at a perturbed revised mean costs at most $4D$ times its error. The record-induced regret is therefore bounded by

$$8D|\beta|\eta_R + 4D\left(C_R\eta_R + \sum_\ell |d_\ell|\eta_\ell\right).$$

The sum bounds errors at a selected port using its unconditional Gaussian marginal. It does not grant unchosen readings to the policy.

In the example, both innovation standard deviations exceed $d = D/2$. The larger-variance minus smaller-variance smoothed benefit decreases with $|m|$. Each reading gain over no reading increases up to d and decreases afterwards. Together with the signs at zero and the crossing, these give precisely the three selection intervals. To verify the difference's monotonicity, differentiate (30). For $0 < m < d$ and $\tau \ge d$,

$$\log\frac{d+m}{d-m} \ge \frac{2m}{d} \ge \frac{2dm}{\tau^2};$$

the derivative of the smoothed benefit in m consequently decreases with τ. For $m \ge d$ the same sign follows directly from the two densities. Rounding the R boundaries with error ϵ_R adds at most $8D|\beta|\epsilon_R$: their competing scores agree at an exact boundary within that distance.

Coefficient and output rounding on a grid of sixteen binary fractional bits contributes mean error at most $(2A+1)2^{-17}$, and hence regret at most $4D(2A+1)2^{-17}$. An actual force displacement error ϵ_D contributes at most $2\epsilon_D(\sqrt{v} + D) + \epsilon_D^2$, because the unforced target has second moment v and the applied displacement is bounded by D. With $B = 16$, $A = 4$, $\epsilon_R = 2^{-21}$ and $\epsilon_D = 10^{-5}$, total regret is at most .000164788. Numerical evaluation gives risk below .097711. A conservative directed-interval calculation also

certifies the strict comparison. Put $s = \sqrt{\mathrm{Var}(m)}$ and $\tau_{\max} = \max_\ell \sqrt{q_\ell}$. On the positive half-line the product of the folded Gaussian density and $\max(0, G_2, G_4)$ is Lipschitz with constant at most

$$\mathcal{L} = \frac{4D\sqrt{2/\pi}}{s} + \frac{2D\tau_{\max}\sqrt{2/\pi}}{s^2}.$$

Use 1024 midpoint intervals on $[0, .3]$ and subtract $\mathcal{L}(.3)^2/(4 \cdot 1024)$ from the gain integral estimate. Dropping its nonnegative remainder beyond .3 preserves a lower bound. Evaluate the finite lattice sums and interval arithmetic at 55 decimal digits. The Gaussian CDF is evaluated with the integrated exponential Taylor polynomial of order 48, using remainder at most $|x|^{99}/(2^{49}49!\,99\sqrt{2\pi})$. For the clipping bound one may use $\overline{\Phi}(z) \le \phi(z)/z$. This gives finite-controller risk below .097763 and fixed-choice risk above .09827985. The certificate concerns this one-round comparison, not an infinite-duration apparatus or formal verification of the interval software.

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